

Assignment 5- Video and real-time analysis

Part 1- Video read/write

We start with reading a video in python and apply some basic tools on it. The video chosen for this oart is from *pixabay.com*. A resource of free video/image and sound clips.

- Read the file “Bangkok.mp4” file using OpenCV. Find the relevant commands to find the video’s metadata and print out following information about the video: Frame Per second (fps), Total number of frames, Height, and Width of the video
- There are many ways to play this file inside google colab. If you just use OpenCV, it will be shown frame by frame. Another approach is to use embedded version using HTML library. For the rest of the steps, it would be easier to use html version for playing the video.
- Let’s apply some filters on the video file. Use `adaptive_theshold_mean` and `threshold_binary_inv`, to create a sketch version of the video. You can use `Videowriter` command to save the output but it takes quite some time. Instead, only print 3 frames of the video.



Part 2- Pose estimation in video

We would like to have a real-time pose estimation. For this part, we will follow the approach in <https://github.com/tensorflow/hub/tree/master/examples> and estimate the pose of the girl in video. A gif file is provided for you, please estimate the pose and show it on the video.

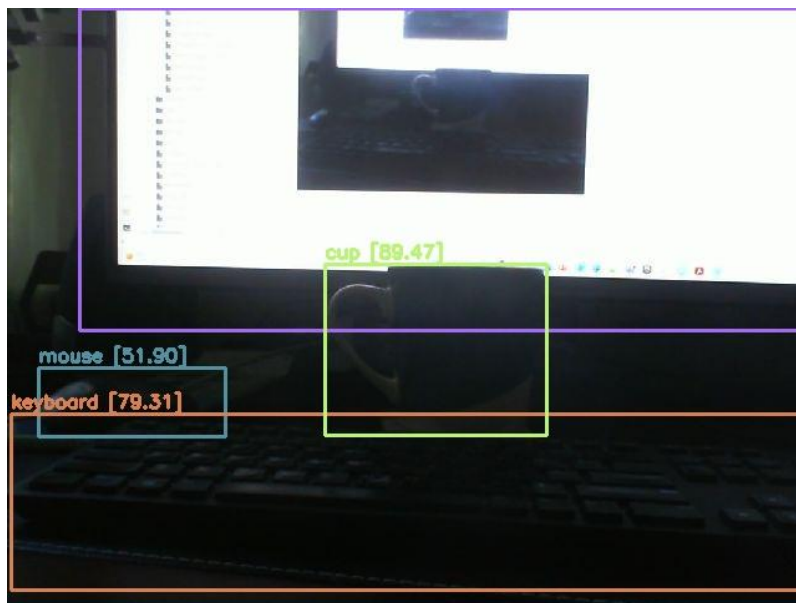
Remember we talked about YOLO earlier. YOLO 26 currently have a tool for pose estimation. Try it out as the next part. Much easier. Right?

Part 3- Using webcam vidoe

Now, we want to use webcam and make object detection on real-time. As you have seen in previous modules, the first step is to have a pre-trained model and use it for future classifications. My recommendation is to use the YOLO11 model for this part (*yolov11n.pt*).

Another piece that we need is a script to connect our webcam to python to make a live object detection application. We will be using the code snippet for Camera Capture which runs JavaScript code to utilize your computer's webcam. Please refer to previous assignment to copy the code here.

Use a code to detect the camera and take a photo and finally detect the objects in the photo.



Extra credit: Design a Real-Time pose estimator.